

3.4.2 P-series Worksheet

Determine whether each of the following series converges or diverges. You may use any theorem from 3.4.2 or earlier.

1. $\sum_{n=1}^{\infty} \left(\frac{2}{n^2} - \frac{2}{2^n} \right)$

We want to try and split this series into two series, but we have to investigate each individual series separately.

Note that $\sum_{n=1}^{\infty} \frac{2}{n^2} = 2 \sum_{n=1}^{\infty} \frac{1}{n^2}$ and the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$ and so this series converges.

Note that $\sum_{n=1}^{\infty} \frac{2}{2^n} = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = \sum_{n=1}^{\infty} 1 \left(\frac{1}{2} \right)^{n-1}$. This is a geometric series with $r = \frac{1}{2}$ and since $|r| < 1$ this series converges.

Since both individual series converge we can say that $\sum_{n=1}^{\infty} \left(\frac{2}{n^2} - \frac{2}{2^n} \right) = \sum_{n=1}^{\infty} \frac{2}{n^2} - \sum_{n=1}^{\infty} \frac{2}{2^n}$ and this series converges. ■

2. $\sum_{n=1}^{\infty} \frac{4^{n-1}}{2^n}$

Note that $\sum_{n=1}^{\infty} \frac{4^{n-1}}{2^n} = \sum_{n=1}^{\infty} \frac{1}{2} \cdot \frac{4^{n-1}}{2^{n-1}} = \sum_{n=1}^{\infty} \frac{1}{2} \cdot \left(\frac{4}{2} \right)^{n-1} = \sum_{n=1}^{\infty} \frac{1}{2} \cdot (2)^{n-1}$. This is a geometric series with $r = 2$ and since $|r| \geq 1$ this series diverges. ■

3. $\sum_{n=1}^{\infty} \frac{\sqrt{n} + 4}{n^2}$

Note that $\sum_{n=1}^{\infty} \frac{\sqrt{n} + 4}{n^2} = \sum_{n=1}^{\infty} \left(\frac{\sqrt{n}}{n^2} + \frac{4}{n^2} \right)$. We want to try and split this series into two series, but we have to investigate each individual series separately.

Note that $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$. This is a p-series with $p = 3/2 > 1$ and so this series converges.

Note that $\sum_{n=1}^{\infty} \frac{4}{n^2} = 4 \sum_{n=1}^{\infty} \frac{1}{n^2}$ and the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$ and so this series converges.

Since both individual series converge we can say that $\sum_{n=1}^{\infty} \frac{\sqrt{n} + 4}{n^2} = \sum_{n=1}^{\infty} \left(\frac{\sqrt{n}}{n^2} + \frac{4}{n^2} \right) = \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2} + \sum_{n=1}^{\infty} \frac{4}{n^2}$ and this series converges. ■

$$4. \sum_{n=1}^{\infty} \frac{1}{n^2 + 4}.$$

First note that this is not a p-series due to the +4 in the denominator of the fraction.

Let $f(x) = \frac{1}{x^2 + 4}$. Note that $f(x)$ is positive, continuous and clearly decreasing on $[1, \infty)$.

$$\text{We then note that } \int_1^{\infty} \frac{1}{x^2 + 4} dx = \lim_{t \rightarrow \infty} \left[\int_1^t \frac{1}{x^2 + 4} dx \right] = \lim_{t \rightarrow \infty} \left[\int_1^t \frac{1}{4 \left(\left(\frac{x}{2} \right)^2 + 1 \right)} dx \right].$$

Let $u = \frac{x}{2}$ and $du = \frac{1}{2} dx$. The bounds become $u(1) = \frac{1}{2}$ & $u(t) = \frac{t}{2}$.

Therefore,

$$\begin{aligned} \lim_{t \rightarrow \infty} \left[\int_1^t \frac{1}{4 \left(\left(\frac{x}{2} \right)^2 + 1 \right)} dx \right] &= \lim_{t \rightarrow \infty} \left[\int_{1/2}^{t/2} \frac{1}{2(u^2 + 1)} du \right] = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \tan^{-1}(u) \right]_{1/2}^{t/2} = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) - \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right) \right] \\ &= \frac{1}{2} \left(\frac{\pi}{2} \right) - \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{4} - \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right) \end{aligned}$$

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$$5. 1 + \frac{2}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \frac{1}{5\sqrt{5}} + \dots$$

We can generalize the terms of this series to express this series as $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$. This is a p-series

with $p = \frac{3}{2} > 1$ and so this series converges.

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$$6. \sum_{k=1}^{\infty} \frac{7}{3k}$$

We note that: $\sum_{k=1}^{\infty} \frac{7}{3k} = \frac{7}{3} \cdot \sum_{k=1}^{\infty} \frac{1}{k}$. Since $\sum_{k=1}^{\infty} \frac{1}{k}$ is the harmonic series (or a p-series with $p = 1$) it diverges and therefore the given series diverges.

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7. $\sum_{i=1}^{\infty} \operatorname{sech}^2(i)$

Let $f(x) = \operatorname{sech}^2(x) = \frac{2}{e^x + e^{-x}}$. We note that f is continuous and positive on $[1, \infty)$. Also,

$$f'(x) = 2 \operatorname{sech}(x) [-\operatorname{sech}(x) \tanh(x)] = -2 \operatorname{sech}^2(x) \tanh(x) = -2 \left(\frac{2}{e^x + e^{-x}} \right)^2 \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) = \frac{-8(e^x - e^{-x})}{(e^x + e^{-x})^3}$$

We see that $f'(x) = 0$ when $x = 0$ and that $f'(x)$ is never undefined. We note that for $x > 0$, $f'(x) < 0$ and therefore we can conclude that f is decreasing on $[1, \infty)$. We then calculate that,

$$\int_1^{\infty} \operatorname{sech}^2(x) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \operatorname{sech}^2(x) dx \right) = \lim_{t \rightarrow \infty} \left(\tanh(x) \Big|_1^t \right) = \lim_{t \rightarrow \infty} (\tanh(t) - \tanh(1)) = 1 - \tanh(1). \text{ This shows us}$$

$\int_1^{\infty} \operatorname{sech}^2(x) dx$ is convergent and therefore, by the integral test $\sum_{i=1}^{\infty} \operatorname{sech}^2(i)$ is convergent. ■

8. $\sum_{k=1}^{\infty} \frac{4k^3 + 2k}{2k^3 - 1}$

Let $a_k = \frac{4k^3 + 2k}{2k^3 - 1}$ and then $f(x) = \frac{4x^3 + 2x}{2x^3 - 1}$. We note that

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{4x^3 + 2x}{2x^3 - 1} = \lim_{x \rightarrow \infty} \frac{4x^3 + 2x}{2x^3 - 1} \cdot \left(\frac{1}{x^3} \right) = \lim_{x \rightarrow \infty} \frac{4 + \frac{2}{x^2}}{2 - \frac{1}{x^3}} = 2 \text{ and thus } \lim_{k \rightarrow \infty} a_k = 2 \text{ Therefore (since the}$$

limit is not equal to 0) the test for divergence states that the given series diverges. ■

9. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$

Let $f(x) = \frac{\ln(x)}{x^2}$. We can see that $f(x)$ is continuous and positive on $[1, \infty)$. We note that

$$f'(x) = \frac{x - 2x \ln(x)}{x^4} = \frac{1 - 2 \ln(x)}{x^3}. \text{ We can see that } f'(x) < 0 \Leftrightarrow 1 - 2 \ln(x) < 0 \Leftrightarrow \frac{1}{2} < \ln(x) \Leftrightarrow e^{1/2} < x.$$

Therefore $f(x)$ is decreasing on $[2, \infty)$. Thus the integral test applies.

$$\int_2^{\infty} \frac{\ln(x)}{x^2} dx = \lim_{t \rightarrow \infty} \left(\int_2^t \frac{\ln(x)}{x^2} dx \right)$$

To continue we use integration by parts.

Let $u = \ln(x)$ and $dv = \frac{1}{x^2} dx$

This implies $du = \frac{1}{x} dx$ and $v = -\frac{1}{x}$.

Therefore we have:

$$\begin{aligned} \int_2^{\infty} \frac{\ln(x)}{x^2} dx &= \lim_{t \rightarrow \infty} \left(\int_2^t \frac{\ln(x)}{x^2} dx \right) = \lim_{t \rightarrow \infty} \left(-\frac{\ln(x)}{x} \Big|_2^t + \int_2^t \frac{1}{x^2} dx \right) = \lim_{t \rightarrow \infty} \left(-\frac{\ln(x)}{x} \Big|_2^t - \frac{1}{x} \Big|_2^t \right) = \lim_{t \rightarrow \infty} \left(-\frac{\ln(t)}{t} + \frac{\ln(2)}{2} - \frac{1}{t} + \frac{1}{2} \right) \\ &= \lim_{t \rightarrow \infty} \left(-\frac{\ln(t)}{t} \right) + \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + \frac{1 + \ln(2)}{2} \right) = \lim_{t \rightarrow \infty} \left(-\frac{\left(\frac{1}{t} \right)}{1} \right) + \frac{1 + \ln(2)}{2} = \frac{1 + \ln(2)}{2} \end{aligned}$$

Therefore, by the integral test, since the integral $\int_2^{\infty} \frac{\ln(x)}{x^2} dx$ converges the series $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^2}$ converges.

So, since our starting series $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$ contains only one additional term it must also converge.

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